

Tactics Techniques and Procedures: 60 Meter Operations

AM-6

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Table of Contents

INTRODUCTION	1
SITUATION / MISSION	1
CONCEPT OF OPERATIONS	2
Activation / Exercises	2
Activation / Actual events	2
Procedures	2
Operating Schedule	2
Role of MARS Stations	3
Broadcasts	3
Infrastructure Assessment Operations	3
Disaster Coordination Operations	4
All Partners Access Network (APAN.org)	4
CONDITONS ON THE USE OF 60 METERS	5
Required Coordination	5
Identification of MARS stations	5
Stations in the RACES	5
Communications between Government and Military Stations	6
General Instructions	6

INTRODUCTION

The five discrete High Frequency (HF) channels in the vicinity of 5 MHz which are assigned to the Amateur Radio Service (ARS) on a secondary basis are authorized to MARS stations on a case by case basis for the limited purposes of establishing initial contact and coordinating with stations operating in the Radio Amateur Civil Emergency Service (RACES). These specific radio frequencies will be referred to generically as “60 meters” through this document.

Channel 1: 5330.5 kHz

Channel 2: 5346.5 kHz

Channel 3: 5357.0 kHz

Channel 4: 5371.5 kHz

Channel 5: 5403.5 kHz

USB Suppressed Carrier (dial) frequencies

SITUATION / MISSION

The United States government has robust communications capabilities that can be used to monitor local conditions and disseminate official information to the public. These capabilities include social media, commercial telecommunications services and news agencies. News agencies in most every community are networked nationally (Associated Press, etc), and broadcast television and radio stations are one of the foundations of the Emergency Alert System (EAS). 60 meter operation, utilizing Amateur Radio operators in local communities as the conduit to those communities, is a measure of last resort and supplements existing capabilities in only the most extreme situations.

The US Military, as well as Federal, State and Local governments have robust multi-level communications plans that cover Primary, Alternate, Contingency and Emergency (PACE) requirements.

It is assumed that through RACES, individual Amateurs Radio Operators and the organizing elements of the ARS such as Community Emergency Response Teams (CERT), American Radio Relay League (ARRL) Amateur Radio Emergency Service (ARES), American Red Cross (ARC), Salvation Army Team Emergency Radio Network (SATERN), etc. are coordinated with local governments and in the most extreme situations are communications conduits to those local governments.

For the purposes of this Plan, RACES refers to operation in the RACES radio service by licensed Amateur Radio Operators that have been authorized by and registered with the appropriate authorities, typically state or county emergency management officials. It is understood that during normal conditions when exercises are conducted, operation will be conducted by licensed Amateur Radio operators and the various organizing elements in the ARS.

It is assumed that Amateur Radio operators possess the skills or abilities that are required to operate in the ARS; however, many may not have specific knowledge or training to perform structured emergency communications tasks. It is also assumed that MARS members possess the skills and abilities required to operate in the ARS, and have specific knowledge and training to perform structured communications tasks.

When other forms of communications are no longer functional, 60 meters operations should perform three primary functions:

- To disseminate official information to the public by communicating information to citizen Amateur Radio operators located in that community.
- To assess conditions and establish situational awareness of local communities by communicating with citizen Amateur Radio operators located in the community.
- To facilitate official federal government and civil affairs communications to local government officials when other PACE capabilities are unavailable.

The priority for 60 meter operations is as follows:

- Federal Emergency Management Agency (FEMA) operations have priority, the Department of Defense is the supporting element to the Department of Homeland Security in domestic emergency preparedness and response activities (domestic natural disasters, etc).
- State and local government operations are coordinated through FEMA.

This plan will outline the organization of 60 meter operations and establish tactics, techniques and procedures (TTP) for 60 meter operation.

CONCEPT OF OPERATIONS

Activation / Exercises

During exercises, the coordination process to use 60 meters is required and 60 meter operation will commence only according to coordinated plans. 60 meter operation will cease at the conclusion of planned exercise activities, or when instructed by exercise authorities.

Activation / Actual events

In cases when:

- a) a national emergency exists,
- b) existing communications and PACE capabilities are impaired,
- c) existing capabilities to monitor local conditions and communicate with the public are impaired,
AND
- d) communications with the coordinating federal government officials does not exist,

60 meter operations MAY commence until such time as these operations are no longer required, OR until the coordination authorities direct that operations are modified or concluded.

Procedures

Procedures and operating modes must be simplified to retain interoperability with all segments of the ARS. The ARS is best described as an ad-hoc communications service made of up licensed, individual citizens of which only a fraction are trained and organized to provide emergency communications (ECOM) services. Of those organized to support ECOM, there is little technical and procedural standardization, with each group and individual adopting technologies and procedures that suits their purposes. The standardization of communications capabilities, procedures and message formats commonly utilized by military and federal stations are varied in the ARS, leaving Single Side Band (SSB) voice as the common platform between all services.

Although some individual amateurs or small groups of amateurs may be capable of adopting procedures to streamline operations; MARS leaders shall resist the temptation to organize amateur activities and instead focus on adapting to the ad-hoc nature of the ARS.

Operating Schedule

The operating schedule and radio frequency plan is intended to present the Amateur Radio community a Keep It Simple Stupid (KISS) approach which is consistent and without operational variables. Operation on the same frequency at the same local time of day is observed at the expense of potential interference and channel congestion.

- During daylight hours, 60 meter operations are relatively short range. Operations are always on Channel 1 at noon local time.
- At night, 60 meter operations are long range. Channel 1 is guarded for calling and operations starting at 2000 hrs local time. After initial contact is made on channel 1, stations that have

continuing communications of long duration may be assigned to channels 2 through 5 as to not interfere with ongoing communications on channel 1.

- Observing local time provides some geographic de-confliction across the continental United States.
- MARS stations operating concurrently on the same frequency will have to operate as parallel net control stations on the same frequency, and coordinate their transmissions to avoid interfering with each other.

The burden of de-conflicting net control stations and maintaining order on the channel rests on the MARS stations that can operate at higher power and have training and experience managing networks of stations in mixed path distances.

Role of MARS Stations

MARS members are authorized in both Federal military service as MARS stations, and in the ARS as licensed individuals. Before becoming MARS stations, MARS members are active Amateur Radio operators. MARS members have a unique insight into both communications environments, and are used as the bridge between these domains.

MARS stations are the primary interface between citizen Amateur Radio operators and federal/military authorities. MARS is organized to execute 60 meter operations when directed, and trains in relevant military and government procedures and message formats. MARS filters the free form information flow and ad-hoc procedure of the ARS into the concise information product in prescribed formats required by military and government leaders.

Broadcasts

Broadcast by high power government and/or military stations are an effective way to disseminate critical information and instructions to both members of the ARS, and through the ARS to the public. Regional or nationwide broadcasts are best conducted after nightfall, when propagation conditions facilitate reception throughout the North America. Previous experience with these broadcasts indicate that high power stations, operating at approximately 1KW with high profile antenna systems, located on the East Coast and West Coast of the United States can effectively be received throughout North America, Hawaii and into Europe.

Previous tests of 60 meter broadcasts indicate that at some locations, the broadcasts are not clearly received due to unusually high levels of locally generated electromagnetic interference (EMI). In most locations, with normal levels of EMI, or where EMI has been mitigated locally, strong reception is reported. During conditions in which these broadcasts would be necessary, it is expected that EMI emitters will be silenced due to the absence of electrical power service, those remaining should be easily located due to the minimal use of electrical equipment.

Government and military stations shall always generate broadcasts on Channel 1.

Infrastructure Assessment Operations

When it is necessary to assess local conditions throughout the United States using Amateur Radio operators, 60 meters is utilized as the HF radio communications interface between Amateurs and government. MARS stations are utilized to contact amateur stations and receive reports of local information. It is always expected these reports are a plain language relation of the local conditions.

MARS operators will receive a request for information that describes the specific information requested. Utilizing MSC software, the request strip becomes a data entry form that ensures information variables are recorded in a consistent manner. The MARS operator will interview the ARS station and prompt for the specific information requested. While it is not necessary or desirable to ask every question presented in the data entry form; MARS operators should avoid generalizing

prompts, such as, “is everything normal in your area?” Instead, use verbal prompts such as “tell me about the status of electrical power in your area” and follow up with specific questions as required.

Disaster Coordination Operations

During natural disasters and other complex catastrophes, Amateur Radio operators are often the first communicators available in the affected area. Because of the dynamic nature of these incidents, it is difficult to precisely plan the communications activities to any great detail.

The following factors should be considered during disaster operations:

- Near Vertical Incidence Skywave (NVIS) communications utilizing basic HF radio is reliable and easily accomplished by experienced amateur personnel in the disaster area up to 150 miles
- Responding federal and military support usually have some HF capabilities
- Initial communications with a citizen in the disaster area is better than no communications or local information resources in the disaster area
- Network connectivity evolves and improves as the incident progresses

While no plan can directly apply to every situation, the concepts in this plan are to: keep it simple, use local people as a conduit to reach local leaders, work with leaders in the Amateur Radio and MARS domains to provide the structure and interface to local amateurs whenever possible.

All Partners Access Network (APAN.org)

The APAN is a DoD computer network that exists on the World Wide Web as a collaboration tool for non-government and non-military partners. Communications resources outside the disaster area utilize the APAN to log information and requests. Defense and Government leaders monitor the APAN, utilize appropriate “hashtags” and discussion groups. MARS stations have access to the APAN. Utilizing 60 meters and Amateur Radio resources, MARS stations can provide national leaders a first look into conditions in the disaster area.

CONDITONS ON THE USE OF 60 METERS

This activity is according to joint authorizations granted to the Federal Emergency Management Agency (FEMA) by the National Telecommunications and Information Administration (NTIA) and is subject to very specific conditions. MARS is listed as a joint authority on these authorizations.

Use of these channels is subject to the following restrictions:

- Authorized emissions are USB Voice (J3B), CW (J2A), PSK (J2B) and RTTY (J2D).
- Store and Forward and other automated operation is not permitted.
- On-Air testing is limited to test message traffic and shall not include training.
- MARS use of these channels must be coordinated between US Army NETCOM and the FEMA spectrum manager with notifications going also to the Federal Communications Commission, the National Telecommunications Information Administration and the American Radio Relay League.
- Government stations are authorized up to 3KW transmitter power.

Required Coordination

MARS members may use these channels as Amateur Radio stations on a secondary basis according to the conditions of their Amateur Radio license and FCC Part 97.

HQ Army MARS is the coordinating agency for MARS' use of the 60 meter channels. MARS members will be notified when use of MARS call signs is approved for exercises or events.

Identification of MARS stations

MARS stations when utilizing these channels as military stations use their MARS station call sign, or their Region's Joint MARS Operating Group (RJMOG)'s assigned tactical call sign (refer to the Global Address Book for tactical call signs). Billet call signs are not appropriate for these operations. When operating in this condition, retain the bearing and demeanor of a radio station operating in US military service. Refer to civilians as Sir or Ma'am, keep verbalizations concise and stay on topic, and display no emotion while operating. Amateur Radio stations custom is to ask for additional identifying, personal information for their log. When asked for your name, simply answer "my call sign is...". When asked for your location answer in the most general terms "South East United States". You may have to repeat these responses for those who are more persistent. "I'm asking for your name, not your call sign", "I understand Sir, my call sign is..."

When operating as an Amateur Radio Station, use your Amateur Radio call sign and conform to Amateur radio customs and practices.

Stations in the RACES

All US Amateur Radio stations are potentially stations operating in the RACES radio service, therefore no distinction is made during ordinary time and exercises. At such time war powers are invoked it is assumed the only operating amateur stations will be in the RACES radio service.

Communications between Government and Military Stations

The primary purpose of these 60 meter operations is interoperability between government/military and RACES stations. Government and military stations have other spectrum to interoperate. Incidental communications between government and military stations are permitted only in the context of interoperability with RACES stations.

General Instructions

Although MARS and Government stations are primary users and stations in the ARS are secondary, we are also professionals. If the channel is occupied by Amateur Radio operators, the appropriate procedure is to break into the conversation, announce that you are a military station and either request the Amateur Radio operators vacate the channel or request their support for your mission.

If Amateur Radio operators refuse to vacate the channel or to assist by continuing their QSO, simply take note of their call signs and move to another channel if possible/available or wait for the QSO to end. Remain courteous, professional and avoid un-necessary confrontation.

Submit the amateur call signs up through the MARS chain of command along with details (dates, times, channel, etc) of the occurrence for possible reporting to authorities.

MARS use of the 60 meter channels will be tracked and reported on a periodic basis to FEMA in order to document MARS use of these resources. All MARS members should complete the 60 Meter Activity Report at www.dodmars.org immediately following your use of these channels.